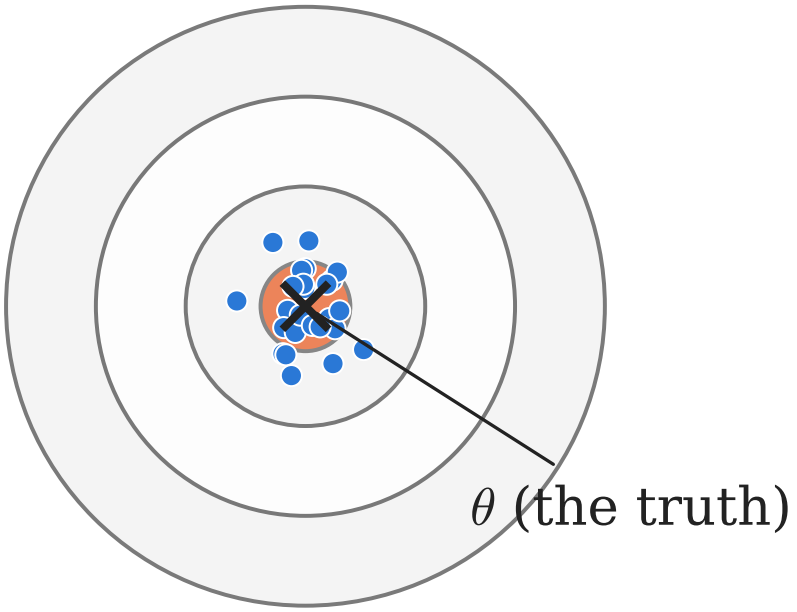
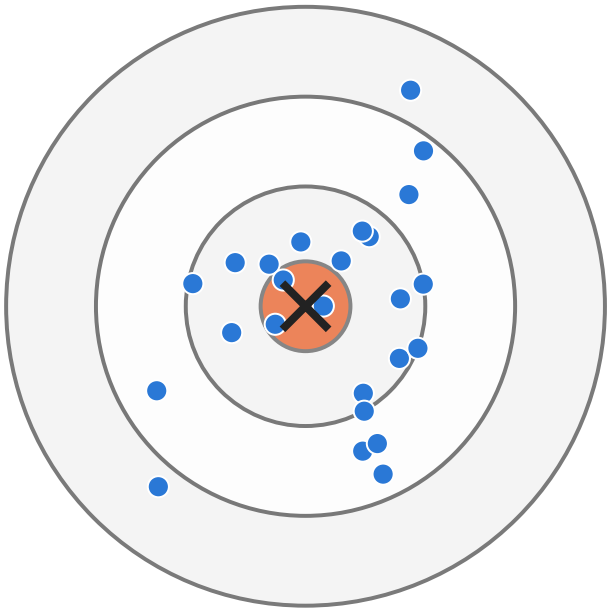
low bias, low variance
(unbiased, precise)



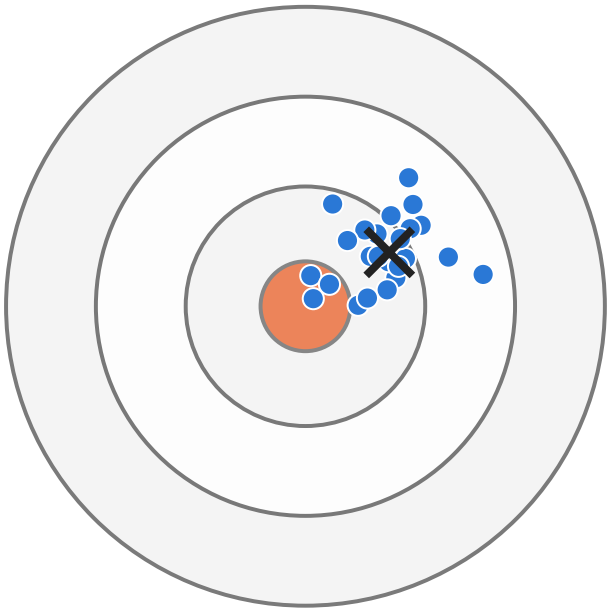
risk $=$ var $+$ bias^2 $= 0.03 + 0.00 = 0.03$

low bias, high variance
(unbiased, imprecise: X_1-like)



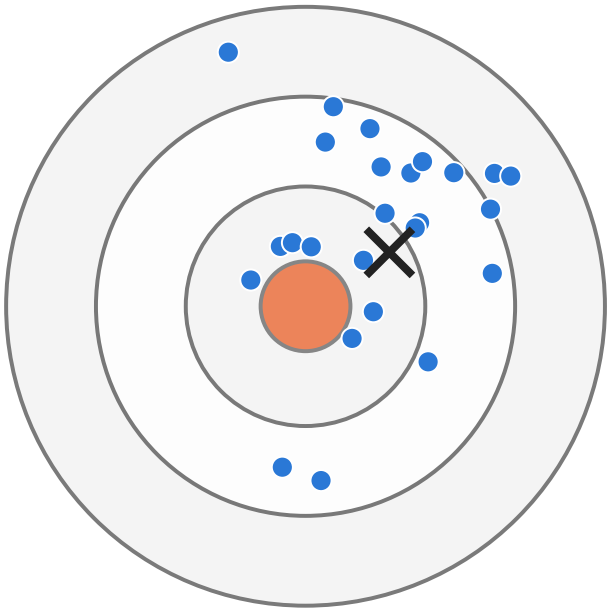
risk $=$ var $+$ bias^2 $= 0.20 + 0.00 = 0.20$

biased, low variance
(the shrinkage estimator)



risk $=$ var $+$ bias^2 $= 0.03 + 0.11 = 0.14$

biased, high variance



risk $=$ var $+$ bias^2 $= 0.20 + 0.11 = 0.32$